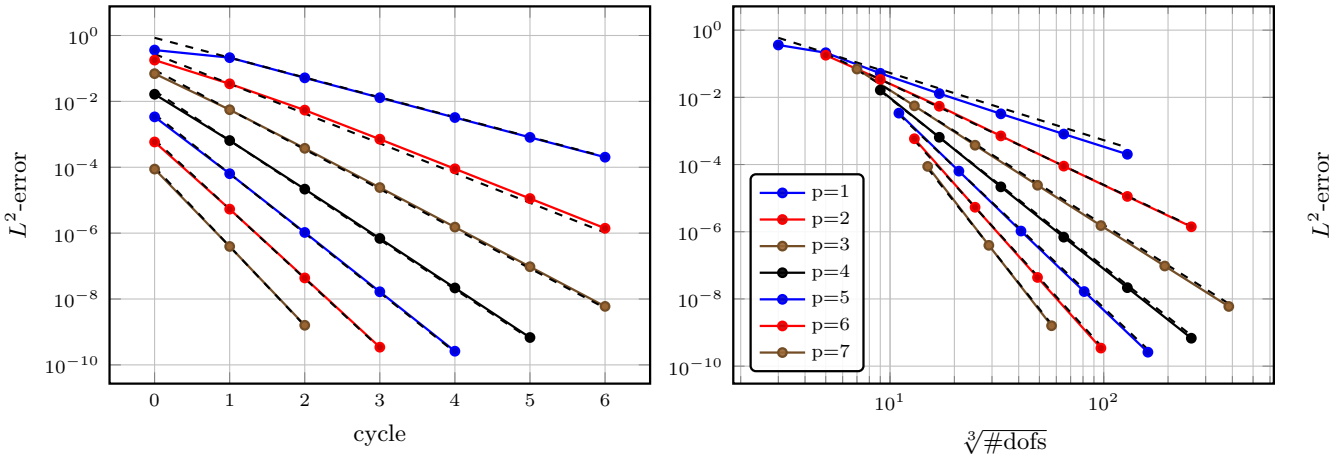
Convergence plots

Each figure compares one non-equidistant node variant with the equidistant node variant. The first row shows the non-equidistant node variant. The second row shows the equidistant node variant. The left column plots the L^2-error over refinement cycle. The right column plots the L^2-error over $\sqrt[3]{\#\text{dofs}}$. The dashed black lines show the expected optimal order.

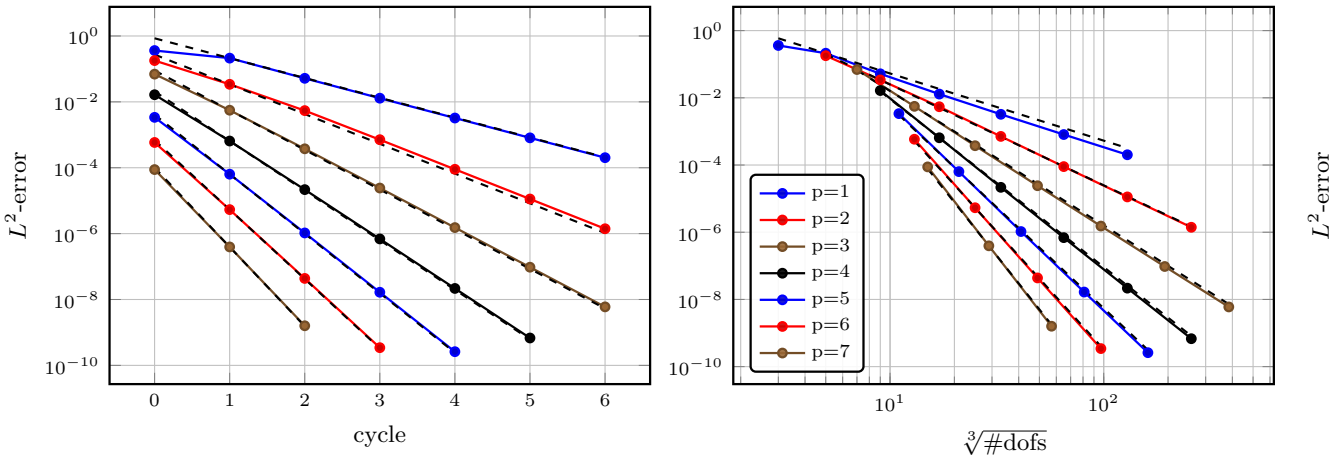
Hexahedron

Continuous Galerkin, Sparse Matrix Vector



(a) Gauss Lobatto: error vs cycle

(b) Gauss Lobatto: error vs $\sqrt[3]{\#\text{dofs}}$

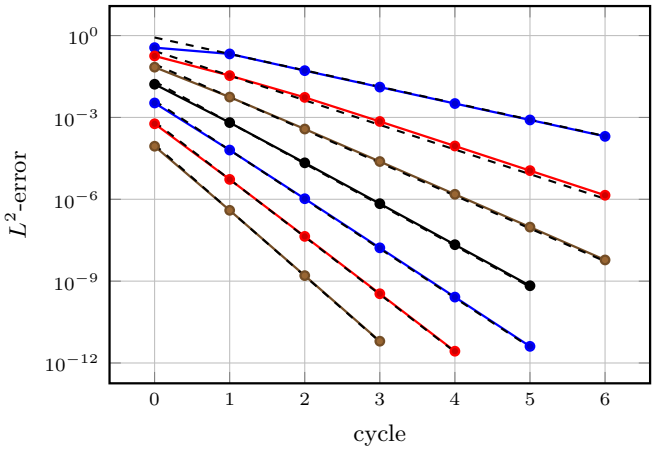


(c) Equidistant: error vs cycle

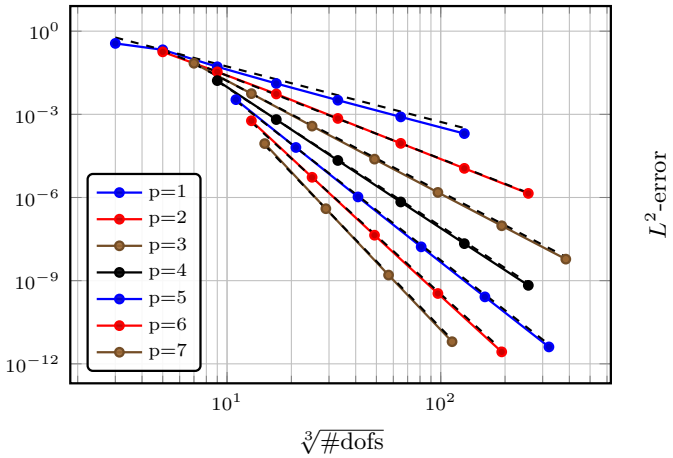
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 1: Hexahedron, Continuous Galerkin, Sparse Matrix Vector: Gauss Lobatto and Equidistant nodes

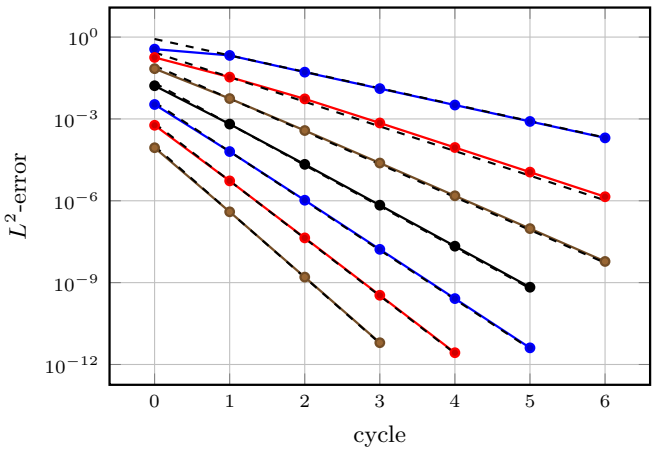
Continuous Galerkin, Matrix-Free



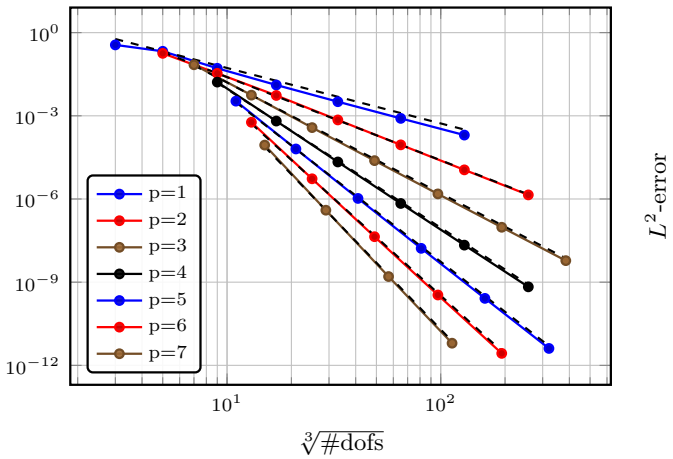
(a) Gauss Lobatto: error vs cycle



(b) Gauss Lobatto: error vs $\sqrt[3]{\#\text{dofs}}$



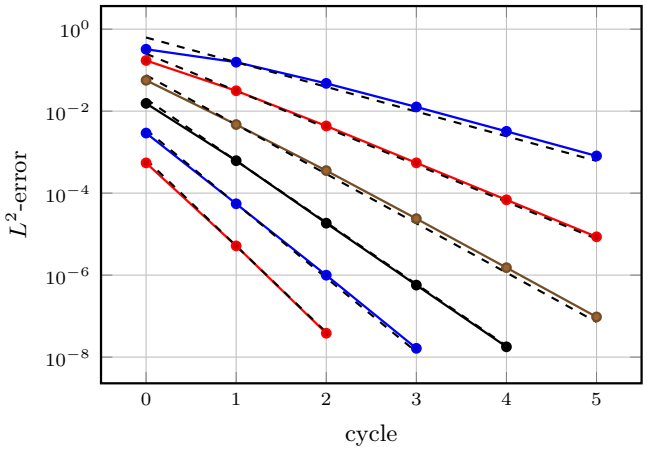
(c) Equidistant: error vs cycle



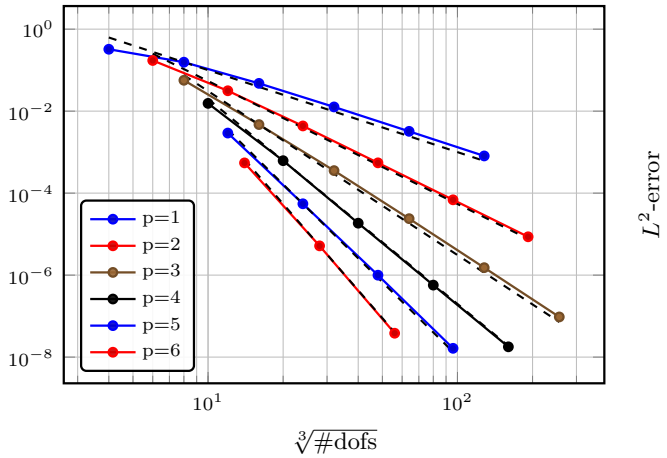
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 2: Hexahedron, Continuous Galerkin, Matrix-Free: Gauss Lobatto and Equidistant nodes

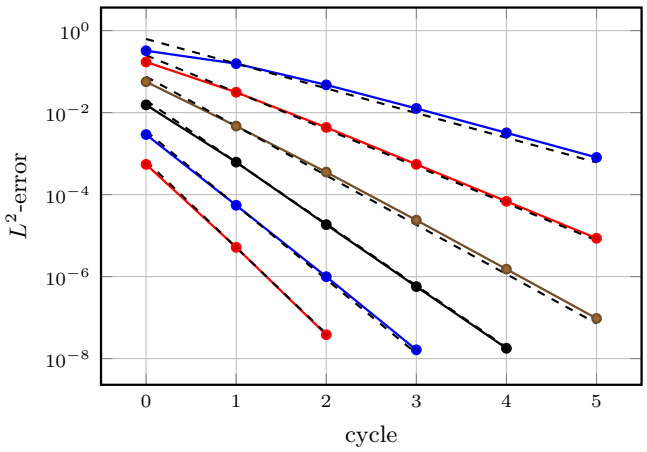
Discontinuous Galerkin, Sparse Matrix Vector



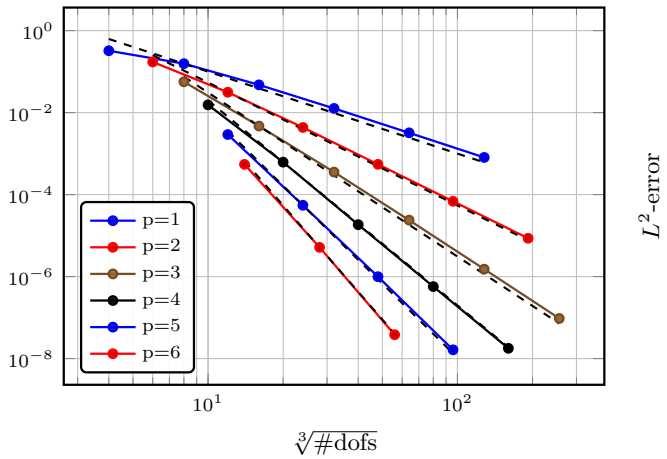
(a) Gauss Lobatto: error vs cycle



(b) Gauss Lobatto: error vs $\sqrt[3]{\#\text{dofs}}$



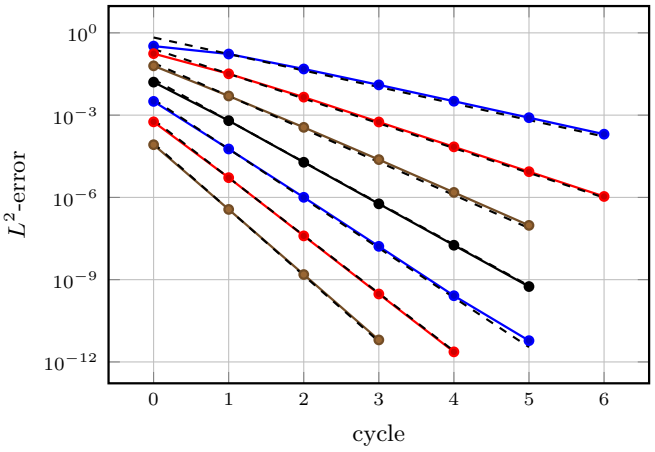
(c) Equidistant: error vs cycle



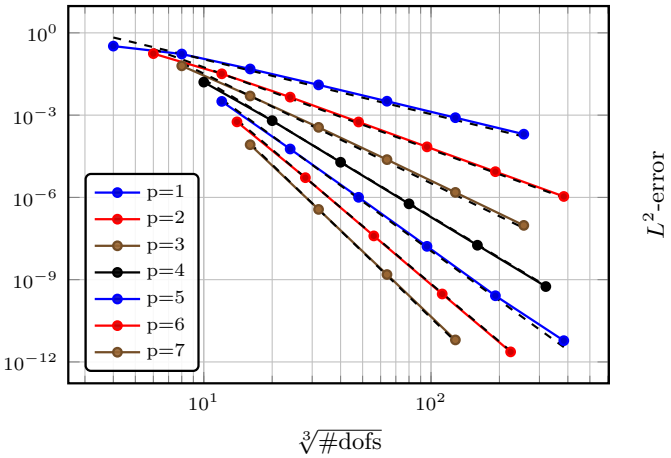
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 3: Hexahedron, Discontinuous Galerkin, Sparse Matrix Vector: Gauss Lobatto and Equidistant nodes

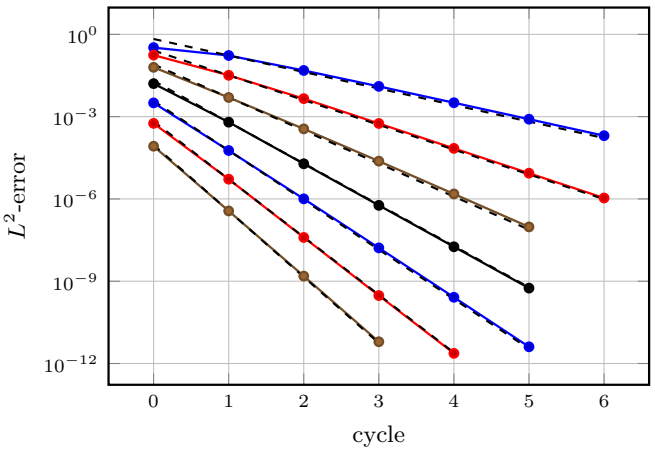
Discontinuous Galerkin, Matrix-Free



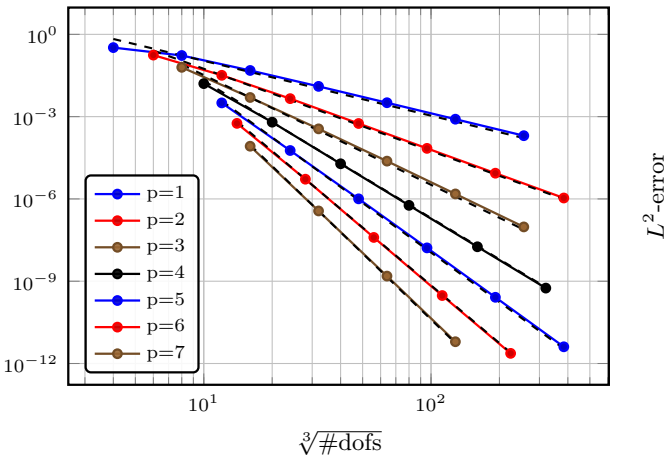
(a) Gauss Lobatto: error vs cycle



(b) Gauss Lobatto: error vs $\sqrt[3]{\#\text{dofs}}$



(c) Equidistant: error vs cycle

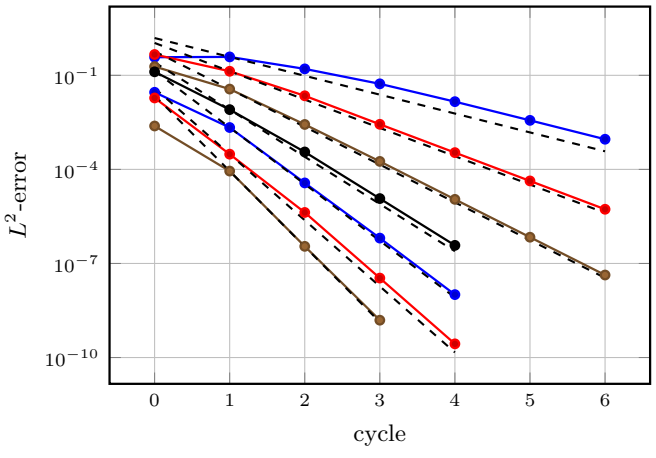


(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

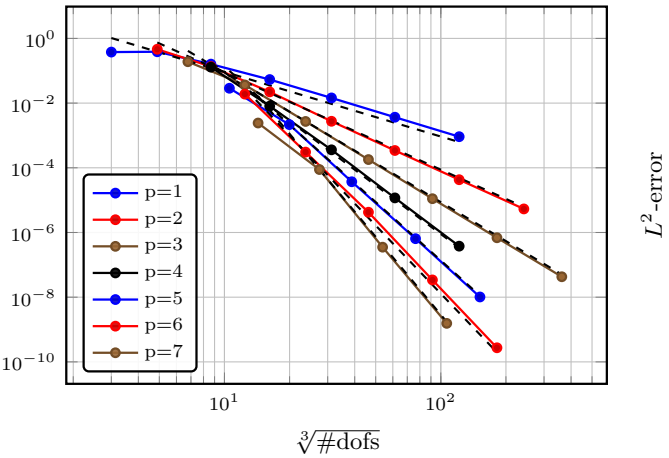
Figure 4: Hexahedron, Discontinuous Galerkin, Matrix-Free: Gauss Lobatto and Equidistant nodes

Tetrahedron

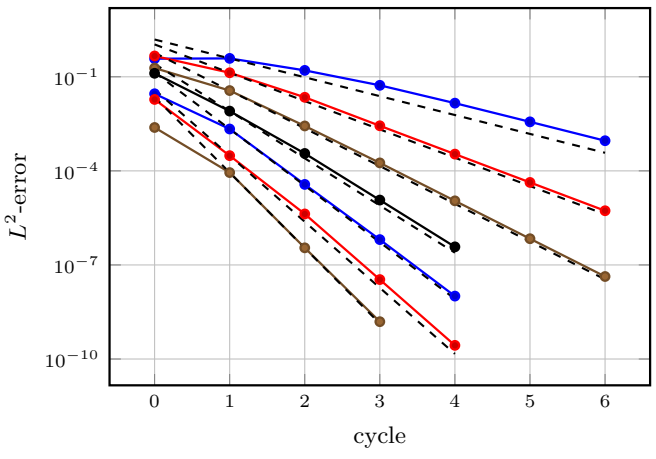
Continuous Galerkin, Sparse Matrix Vector



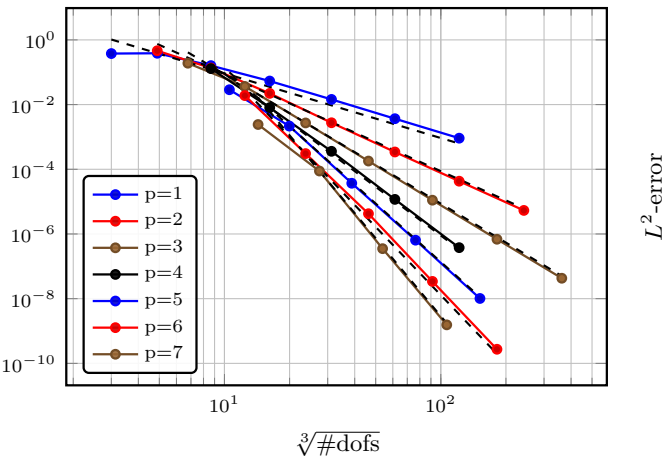
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



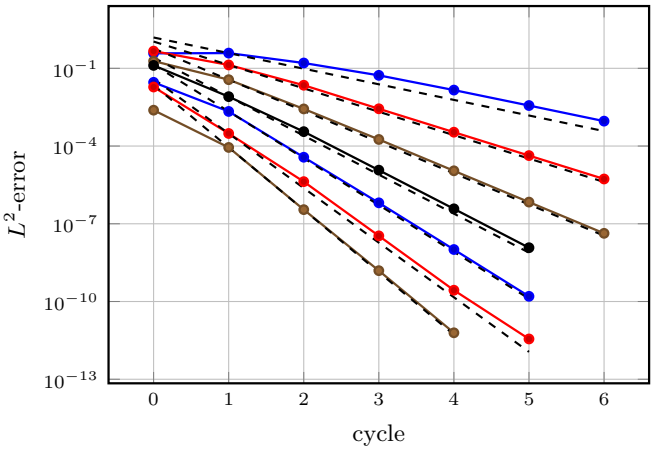
(c) Equidistant: error vs cycle



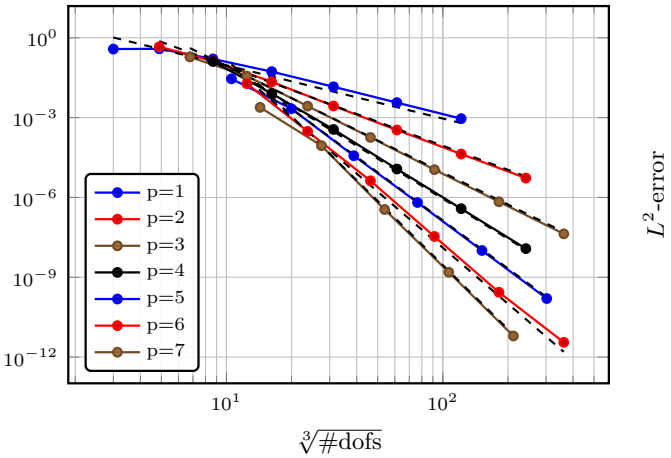
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 5: Tetrahedron, Continuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

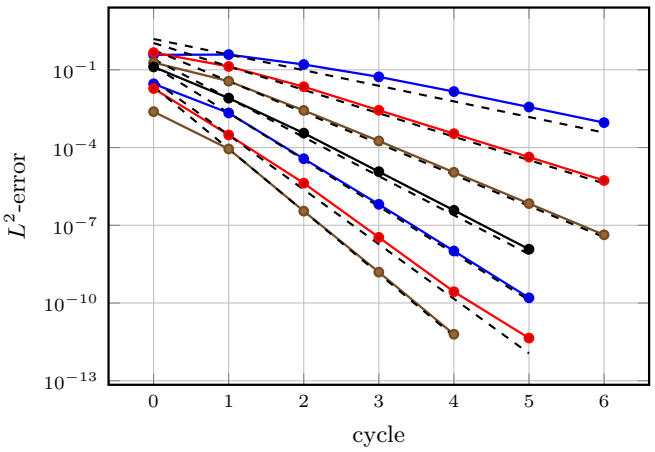
Continuous Galerkin, Matrix-Free



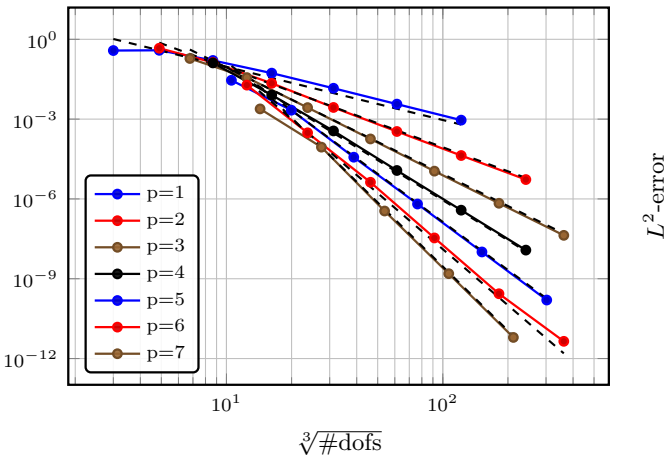
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



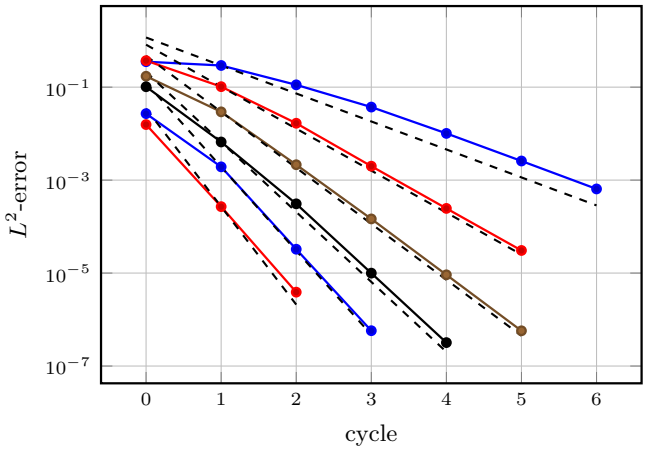
(c) Equidistant: error vs cycle



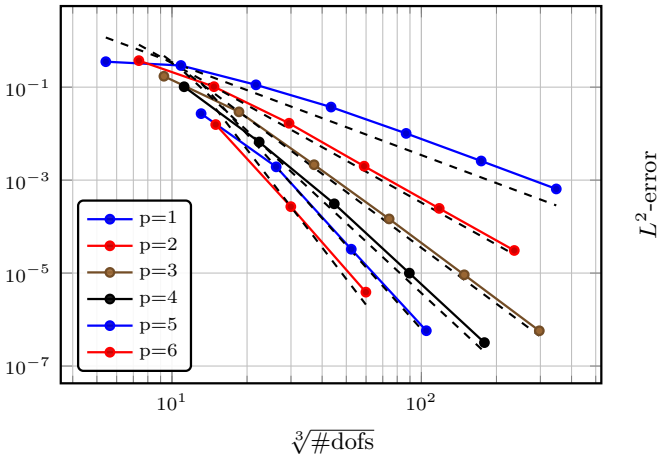
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 6: Tetrahedron, Continuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

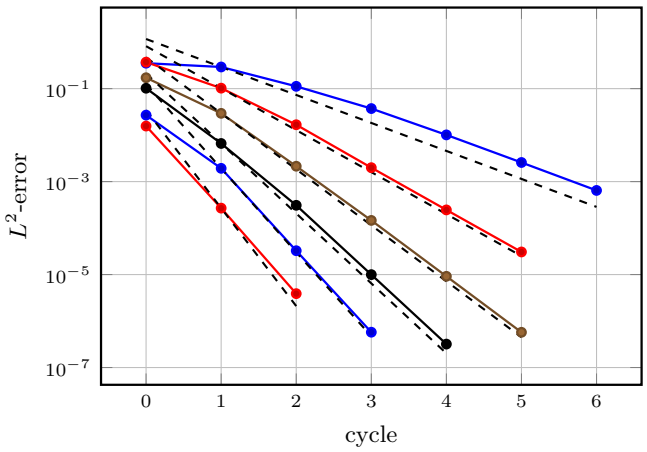
Discontinuous Galerkin, Sparse Matrix Vector



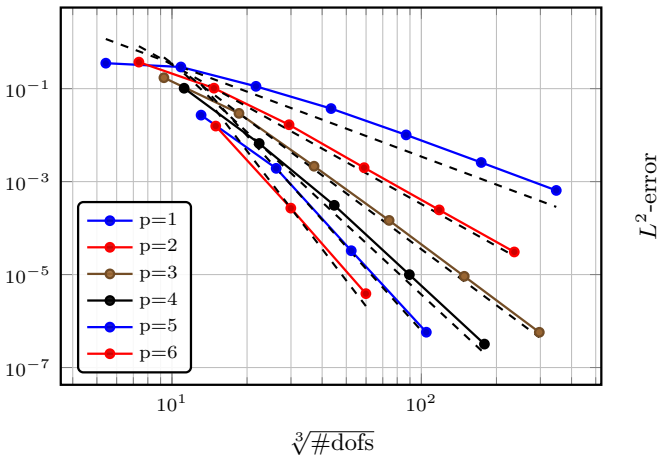
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



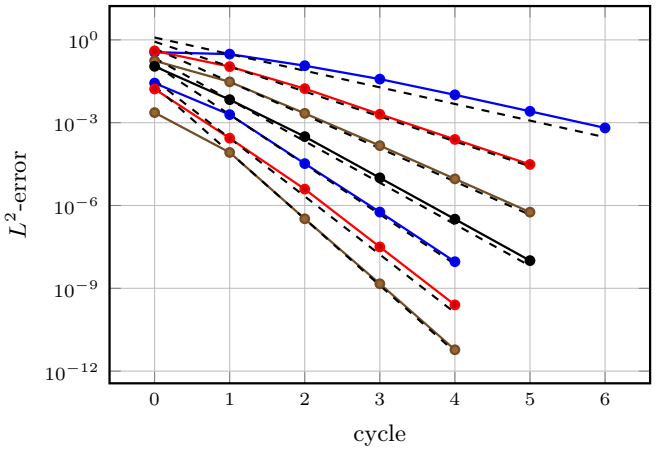
(c) Equidistant: error vs cycle



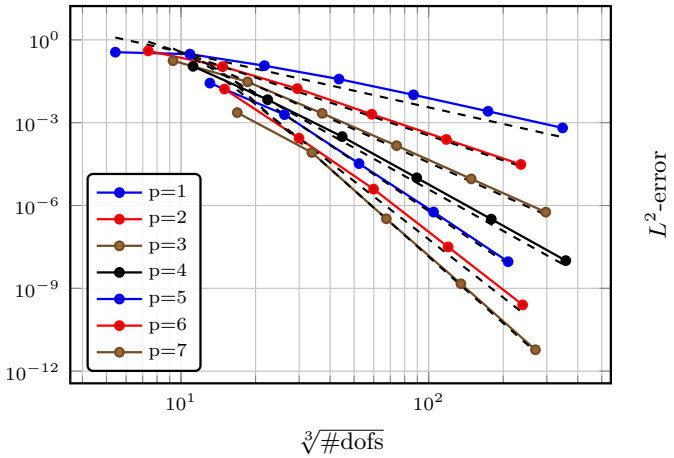
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 7: Tetrahedron, Discontinuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

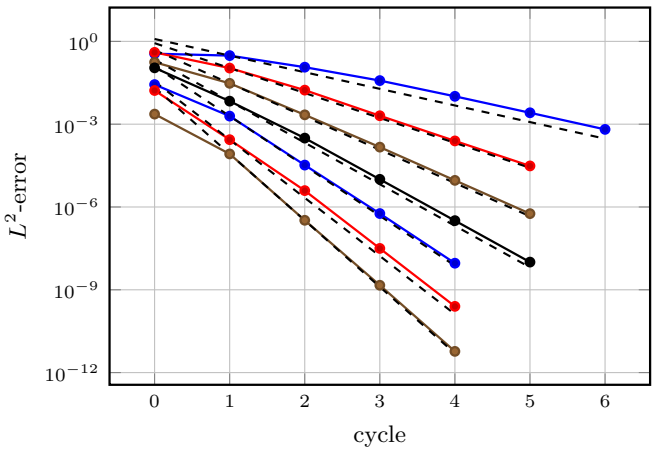
Discontinuous Galerkin, Matrix-Free



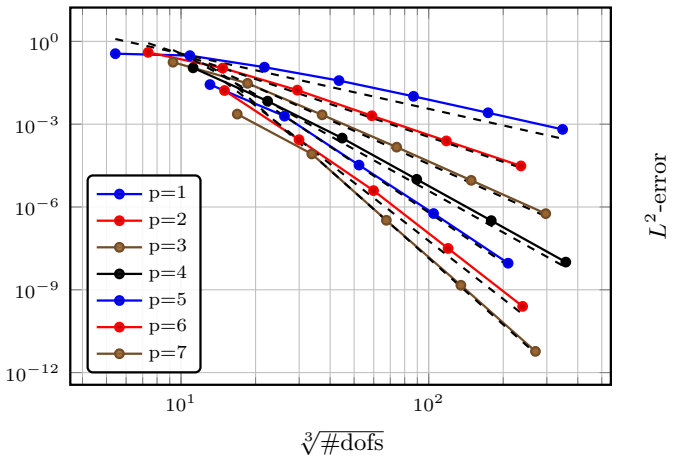
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



(c) Equidistant: error vs cycle

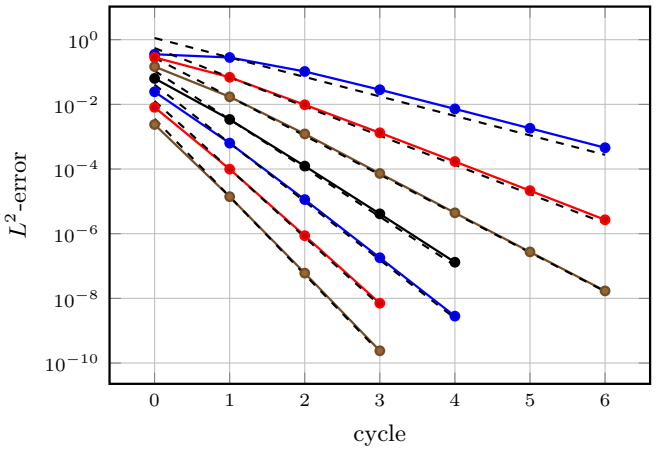


(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

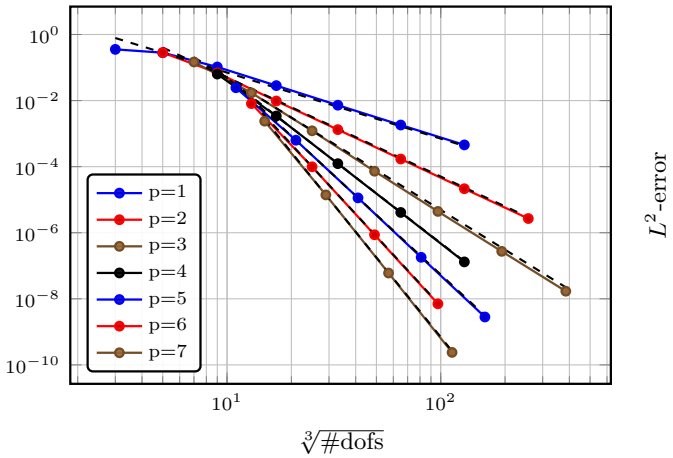
Figure 8: Tetrahedron, Discontinuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

Wedge

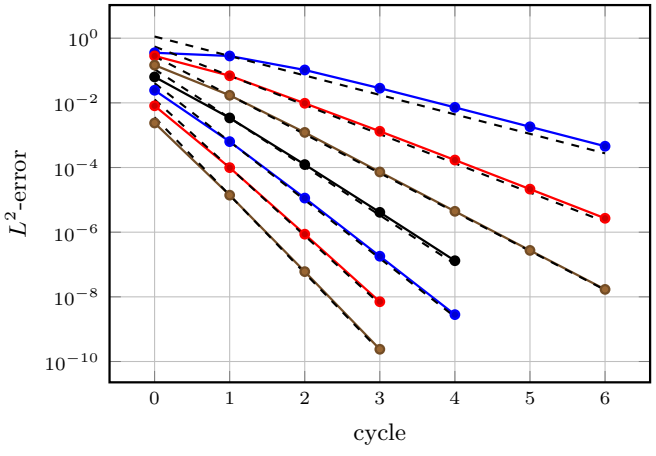
Continuous Galerkin, Sparse Matrix Vector



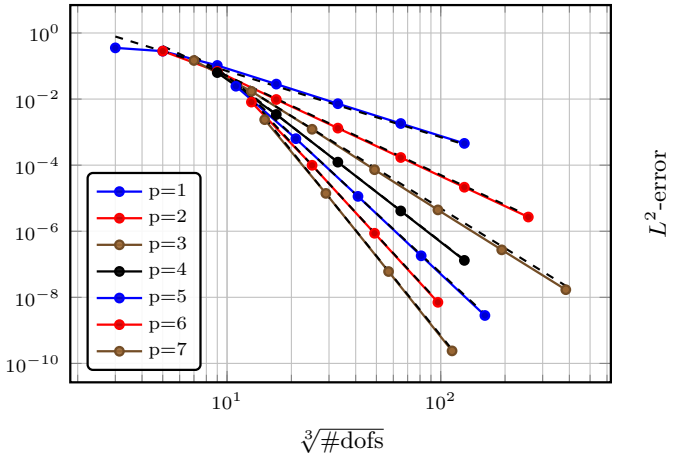
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



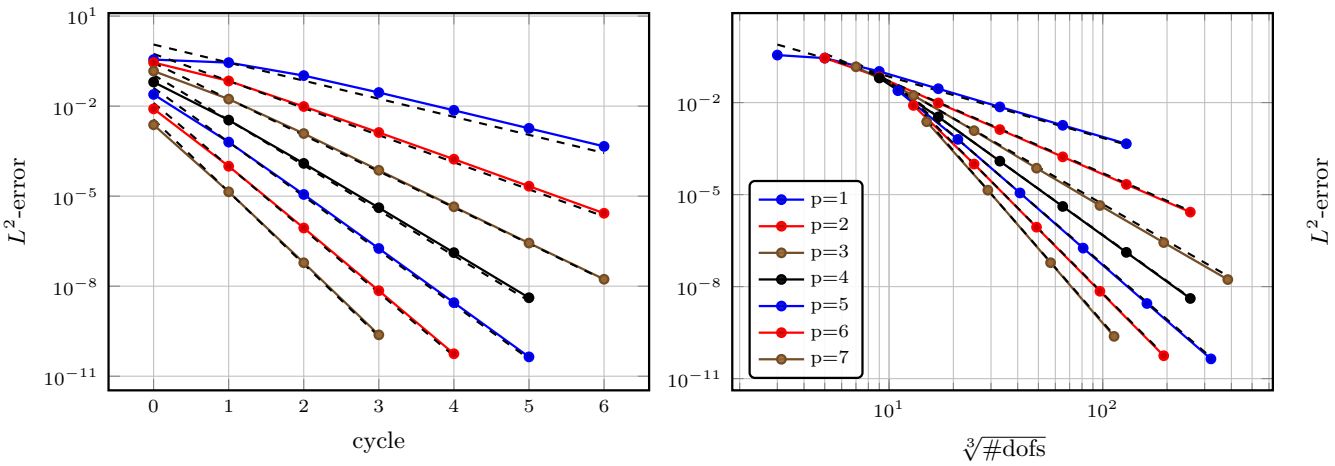
(c) Equidistant: error vs cycle



(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

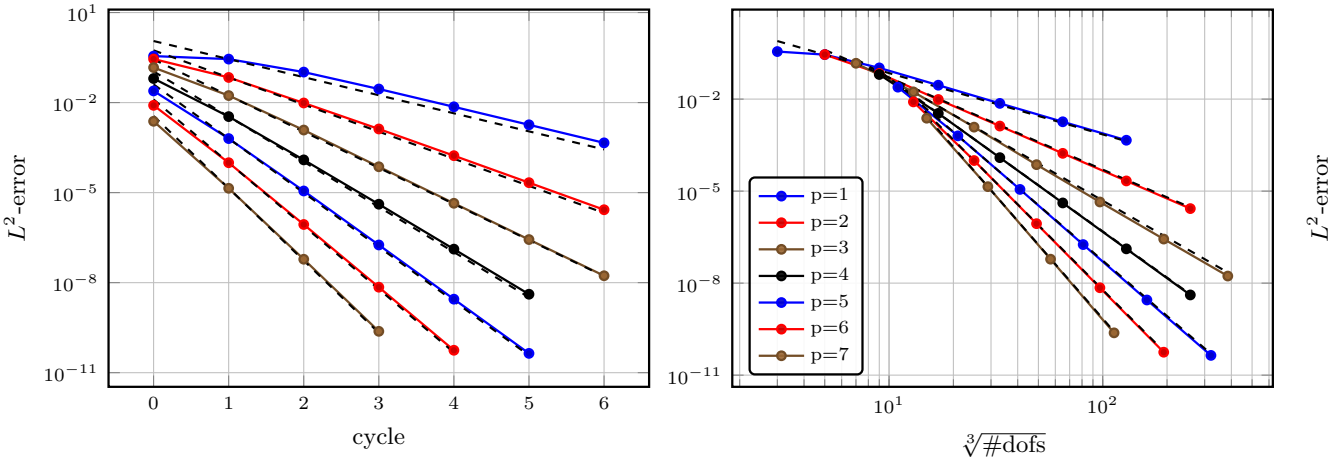
Figure 9: Wedge, Continuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

Continuous Galerkin, Matrix-Free



(a) Blend and Warp: error vs cycle

(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$

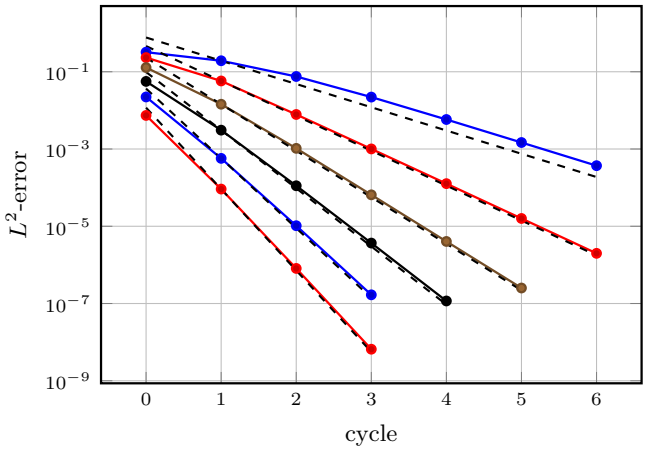


(c) Equidistant: error vs cycle

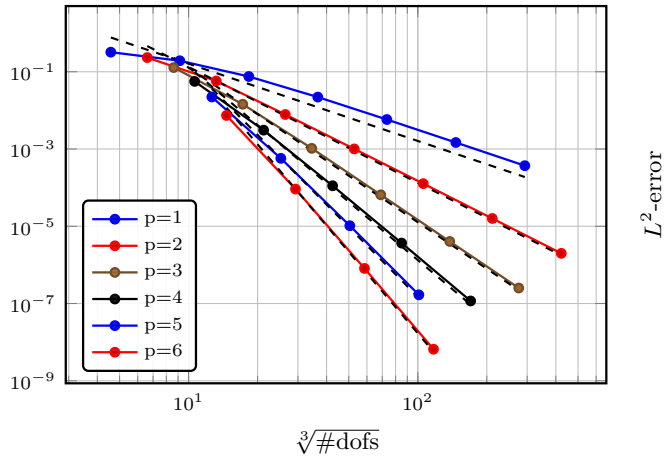
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 10: Wedge, Continuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

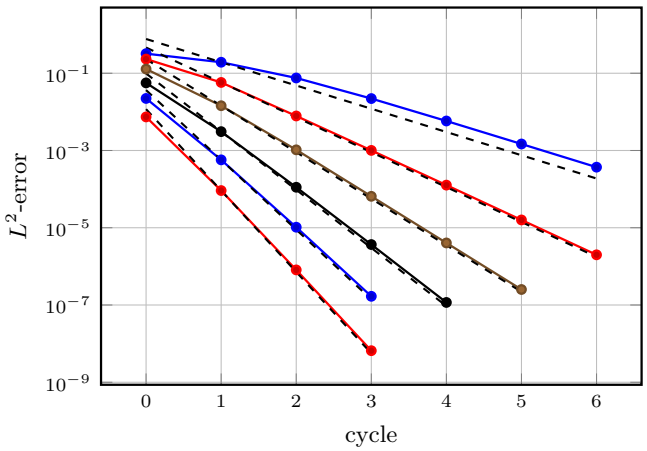
Discontinuous Galerkin, Sparse Matrix Vector



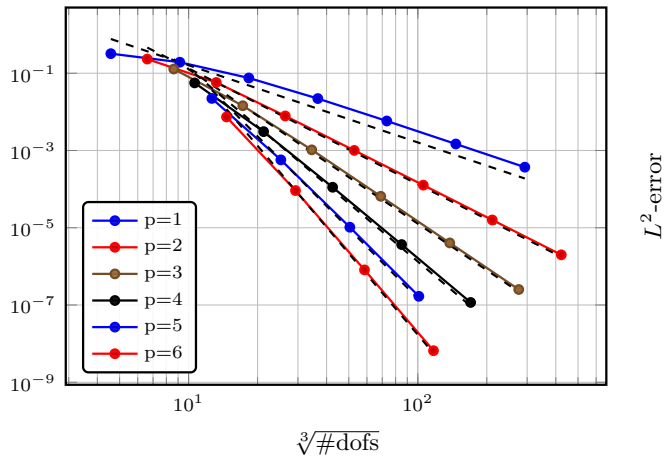
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



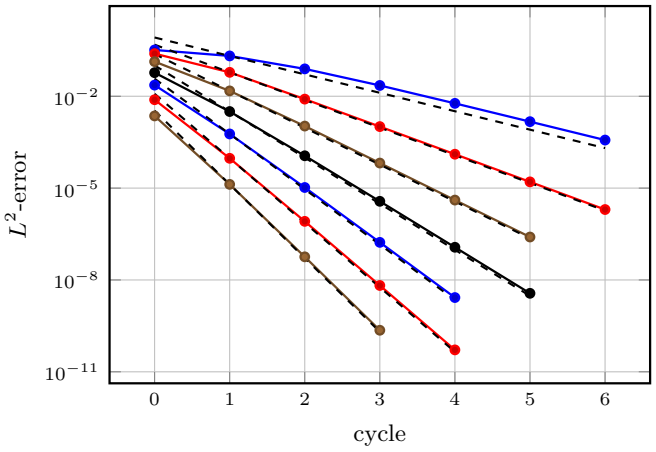
(c) Equidistant: error vs cycle



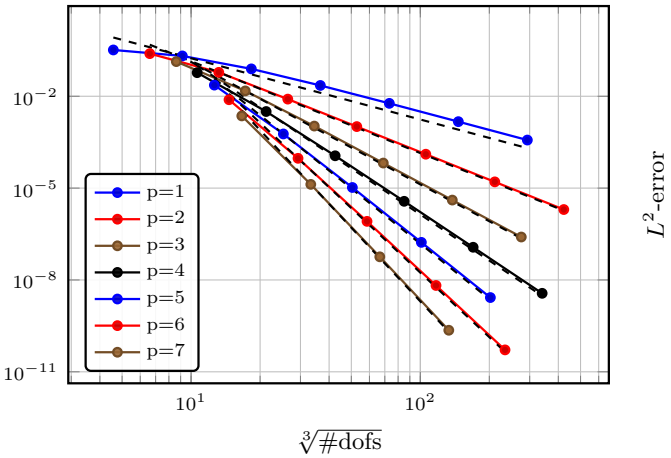
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 11: Wedge, Discontinuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

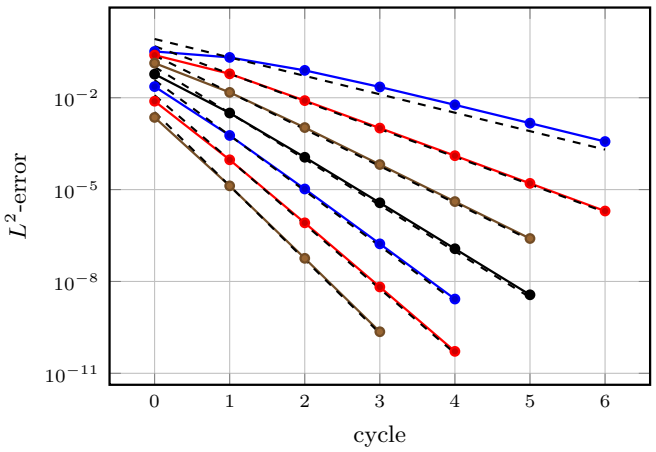
Discontinuous Galerkin, Matrix-Free



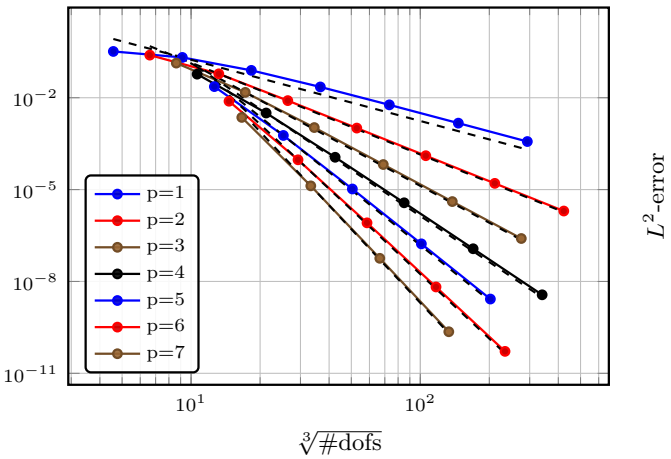
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



(c) Equidistant: error vs cycle

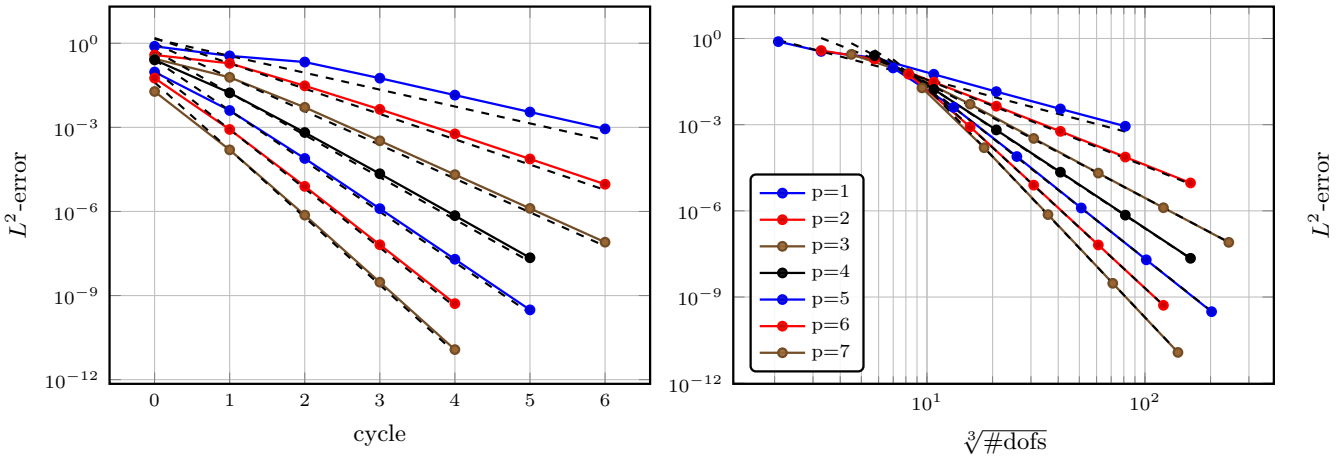


(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 12: Wedge, Discontinuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

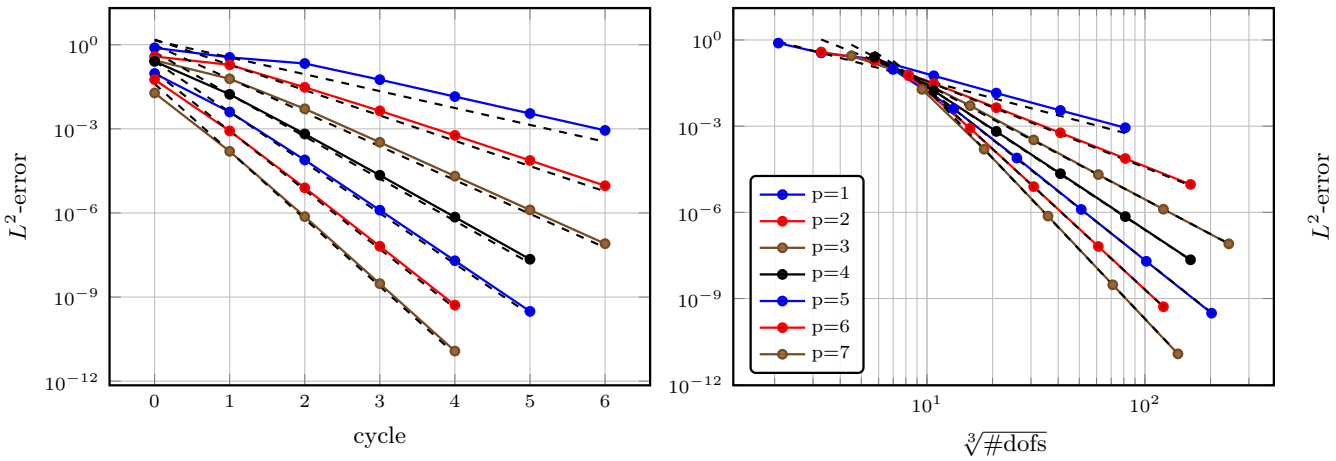
Pyramid

Continuous Galerkin, Sparse Matrix Vector



(a) Blend and Warp: error vs cycle

(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$

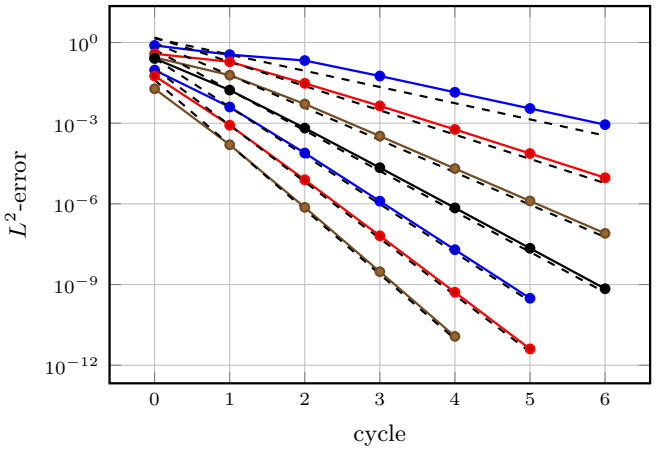


(c) Equidistant: error vs cycle

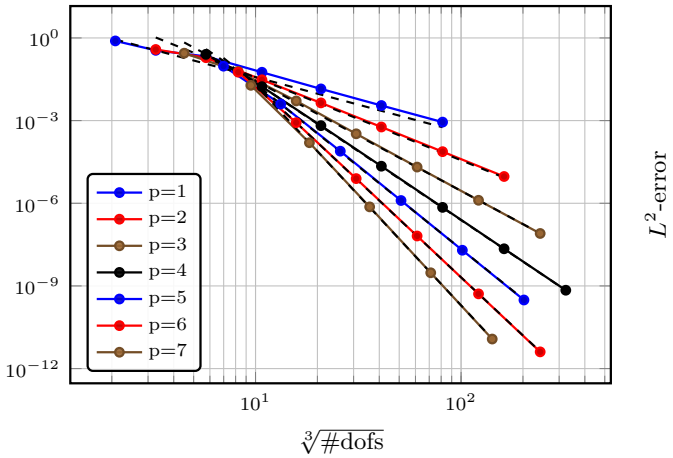
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 13: Pyramid, Continuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

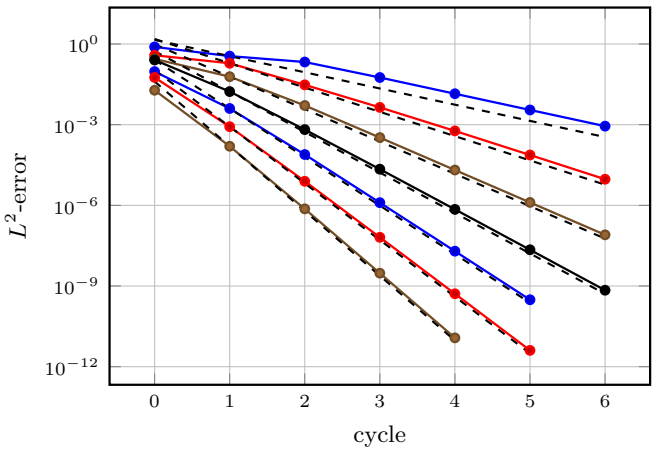
Continuous Galerkin, Matrix-Free



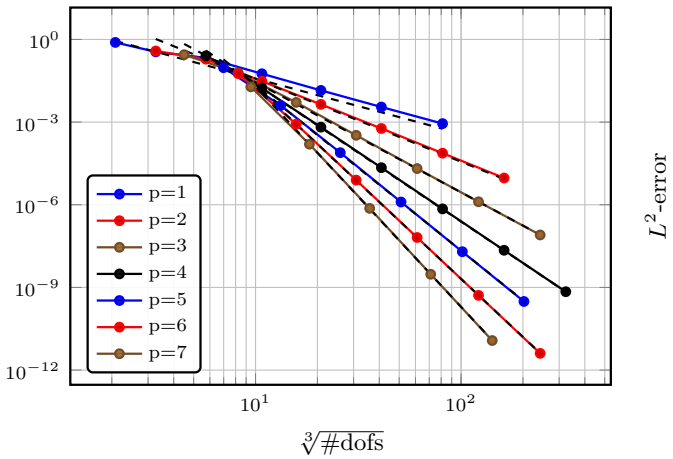
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



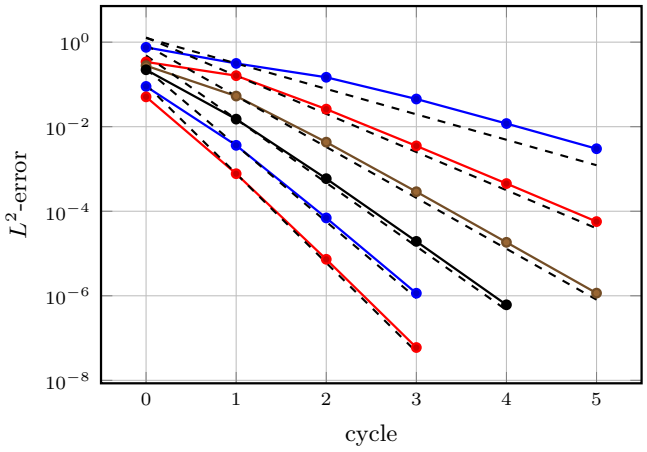
(c) Equidistant: error vs cycle



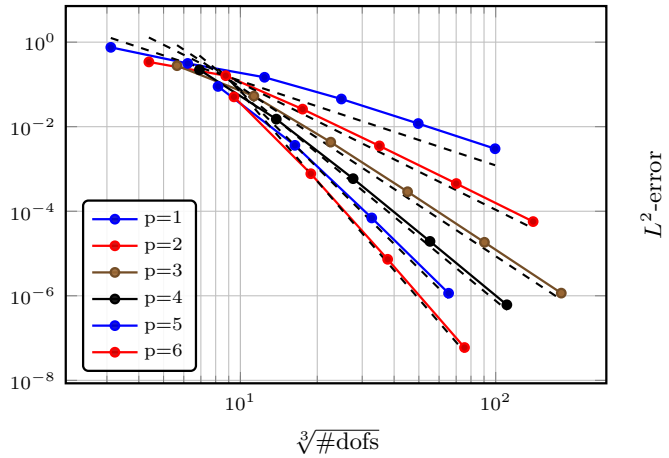
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 14: Pyramid, Continuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

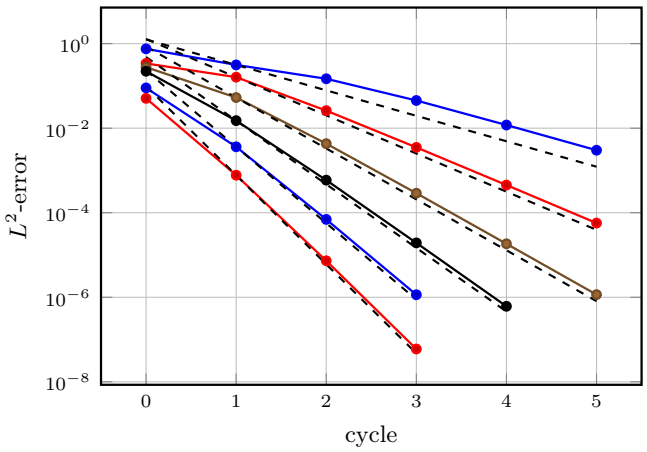
Discontinuous Galerkin, Sparse Matrix Vector



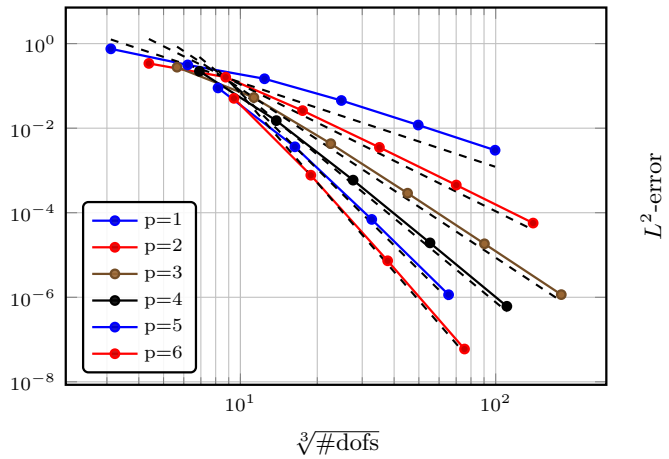
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



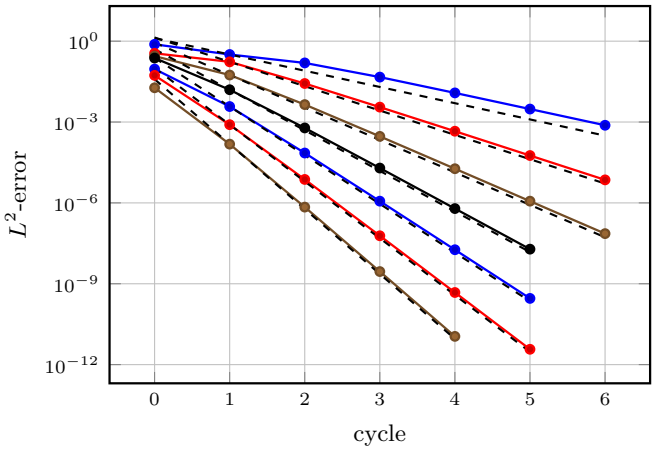
(c) Equidistant: error vs cycle



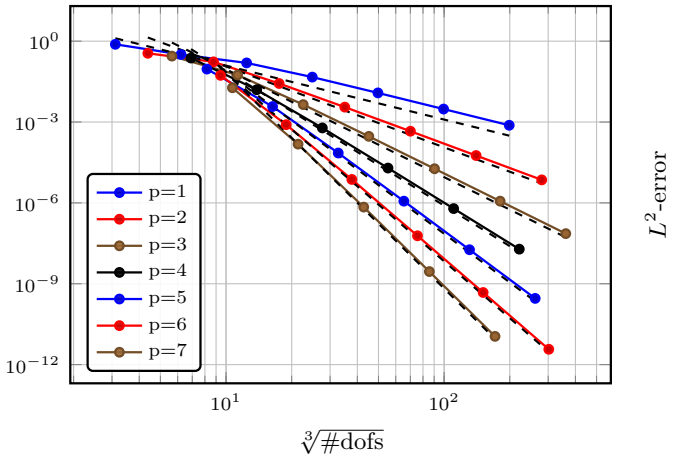
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 15: Pyramid, Discontinuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

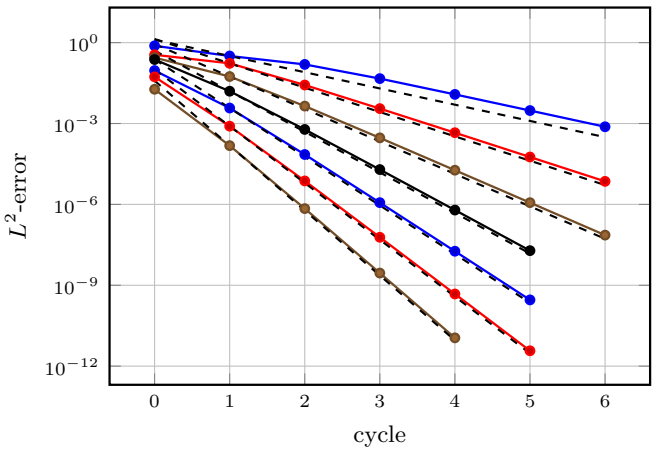
Discontinuous Galerkin, Matrix-Free



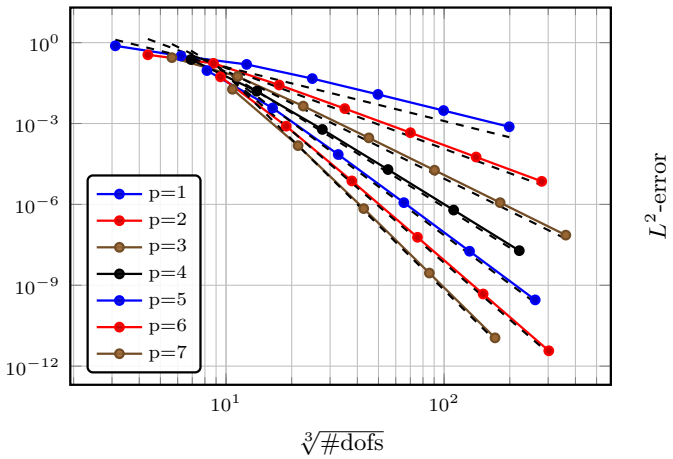
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



(c) Equidistant: error vs cycle

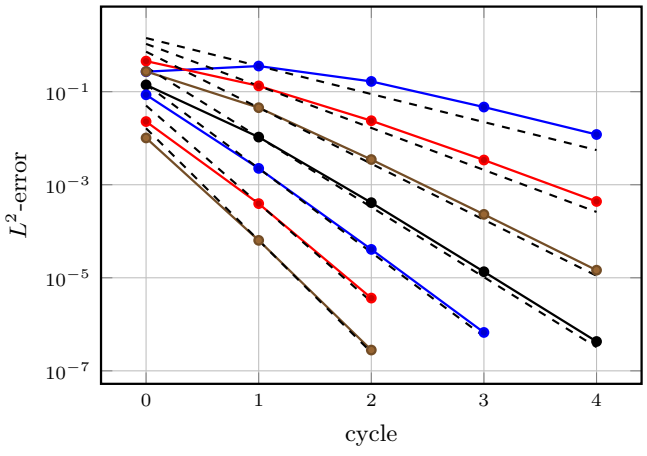


(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

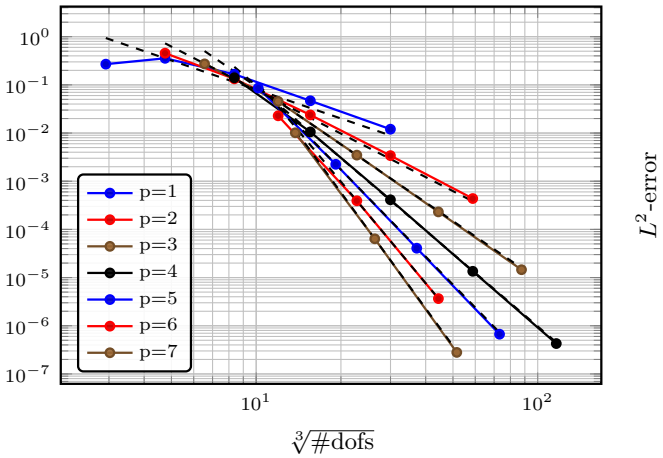
Figure 16: Pyramid, Discontinuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

Mixed

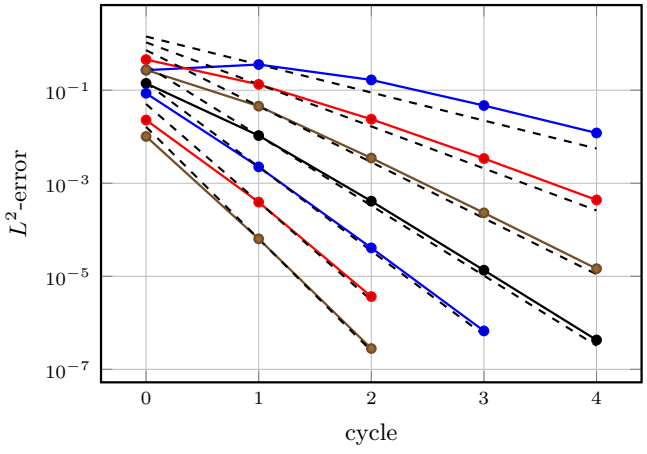
Continuous Galerkin, Sparse Matrix Vector



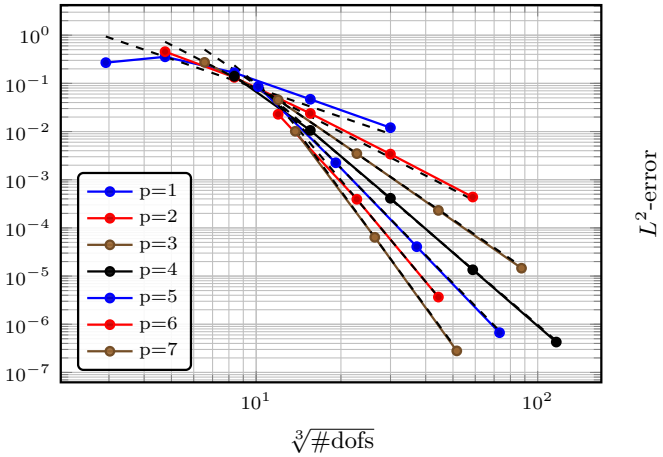
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



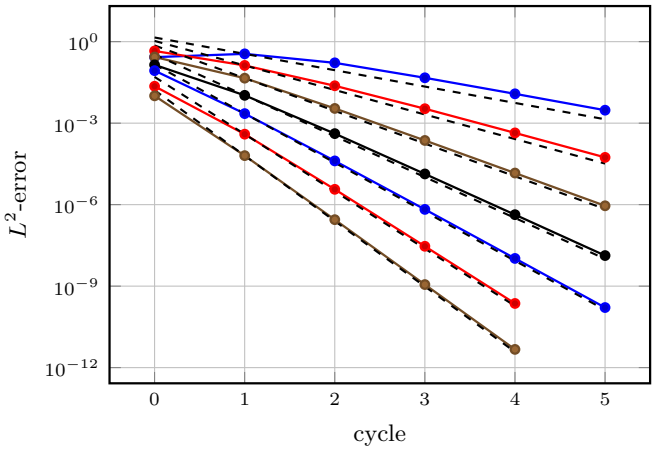
(c) Equidistant: error vs cycle



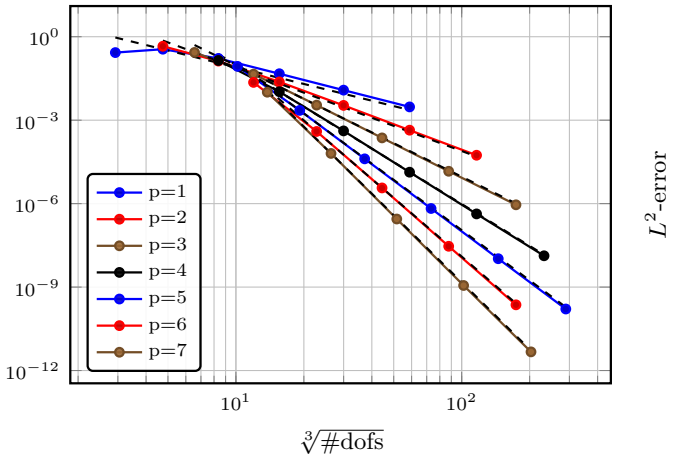
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 17: Mixed, Continuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

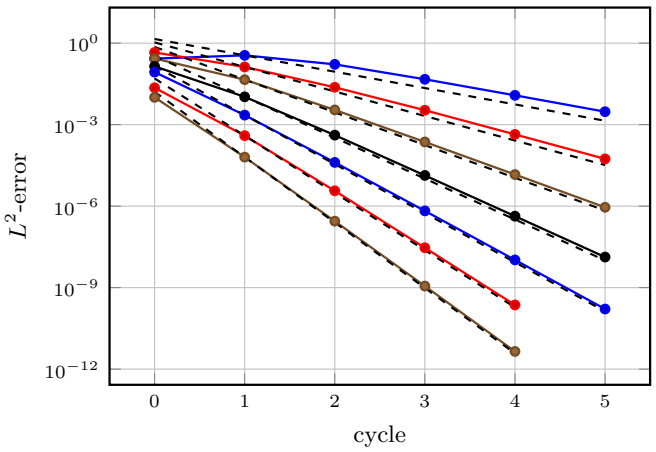
Continuous Galerkin, Matrix-Free



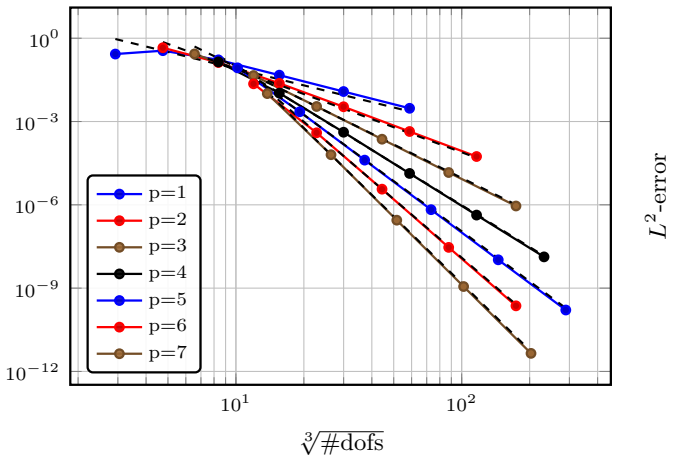
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



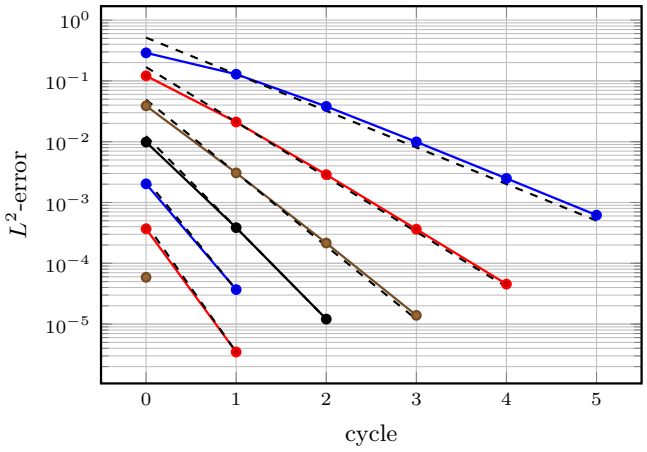
(c) Equidistant: error vs cycle



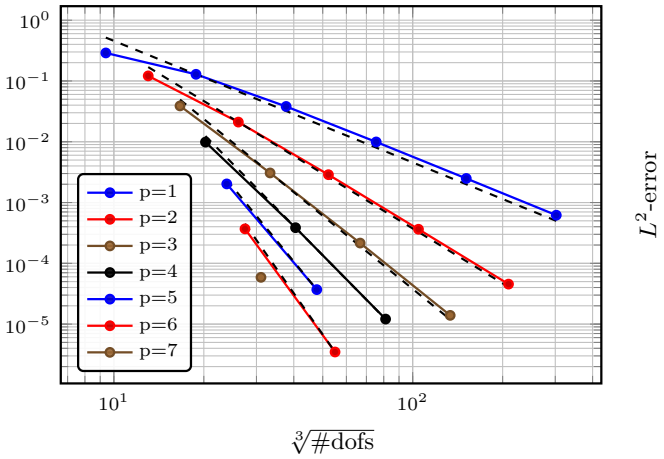
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 18: Mixed, Continuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

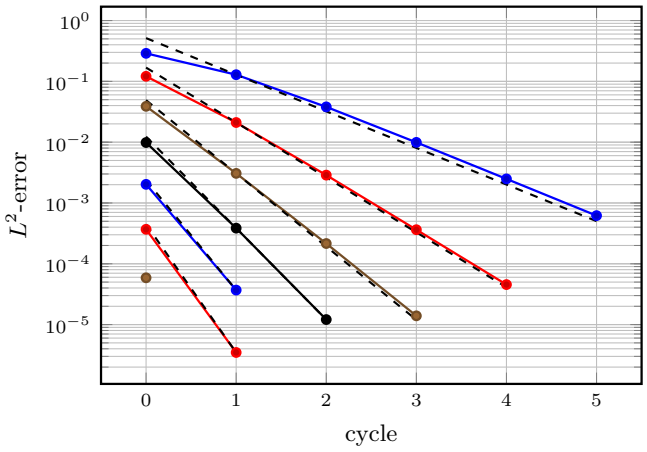
Discontinuous Galerkin, Sparse Matrix Vector



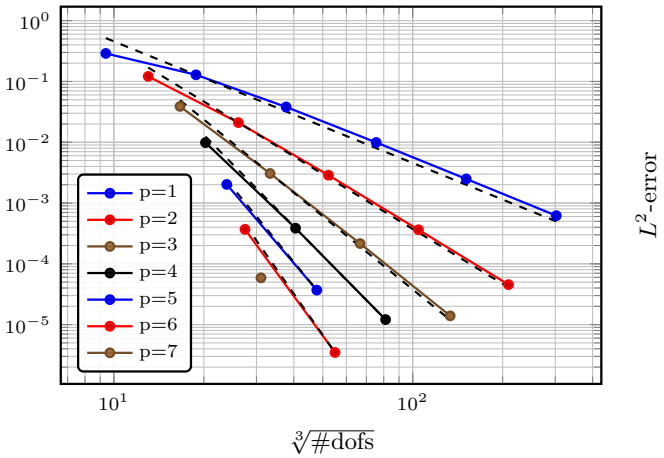
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



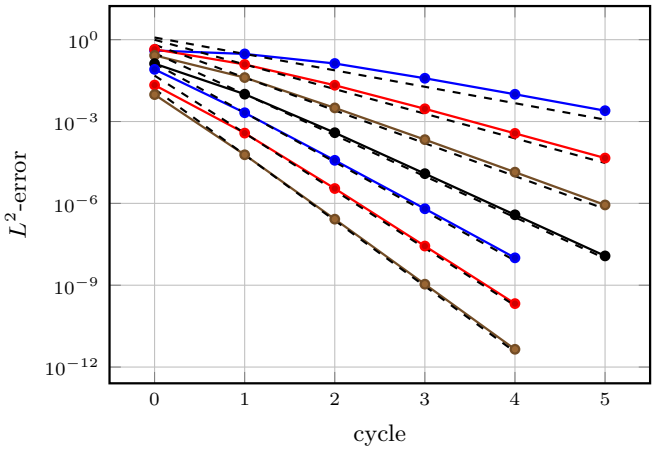
(c) Equidistant: error vs cycle



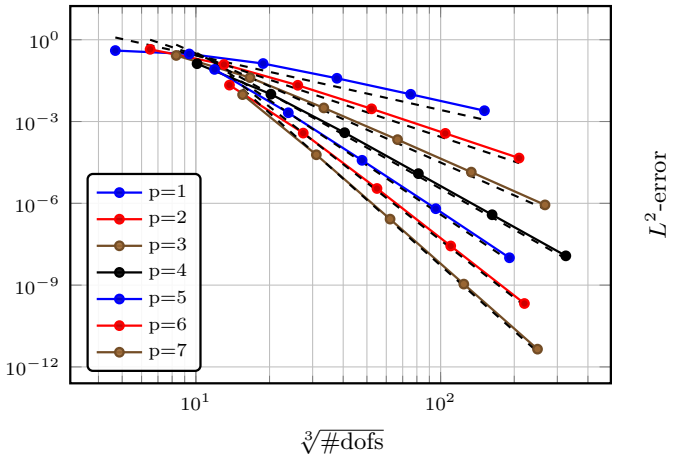
(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 19: Mixed, Discontinuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

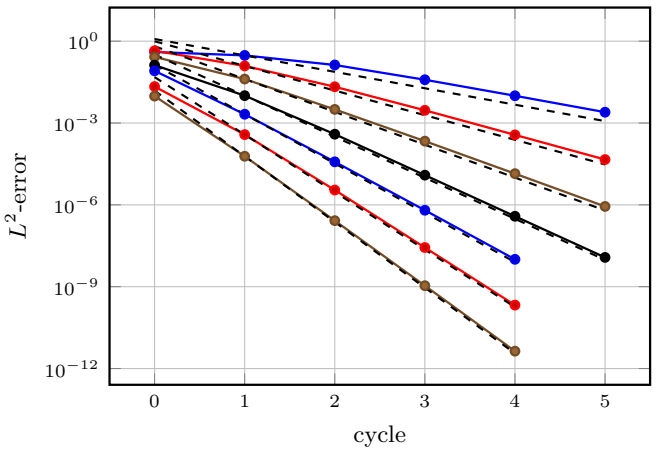
Discontinuous Galerkin, Matrix-Free



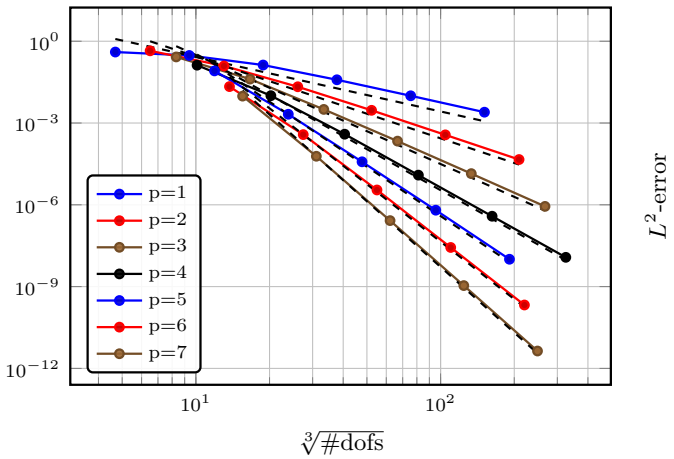
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



(c) Equidistant: error vs cycle

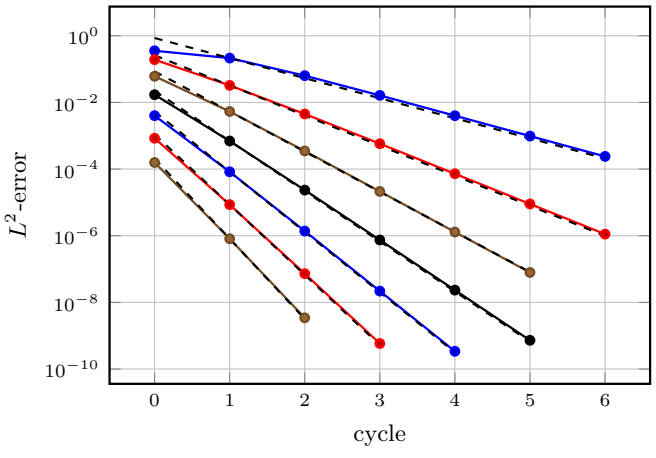


(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

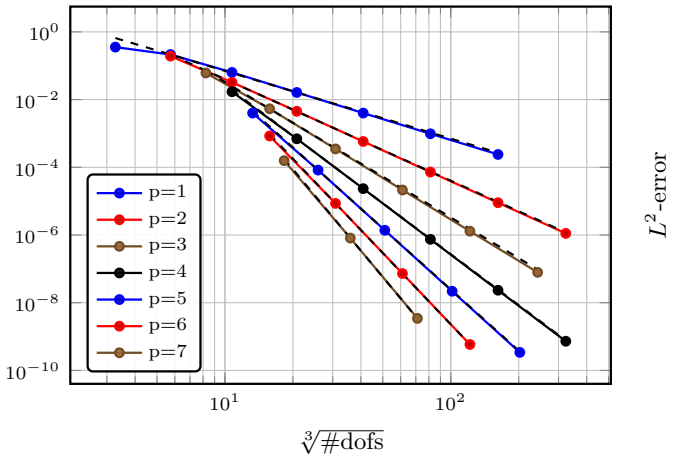
Figure 20: Mixed, Discontinuous Galerkin, Matrix-Free: Blend and Warp and Equidistant nodes

MixedPyramid

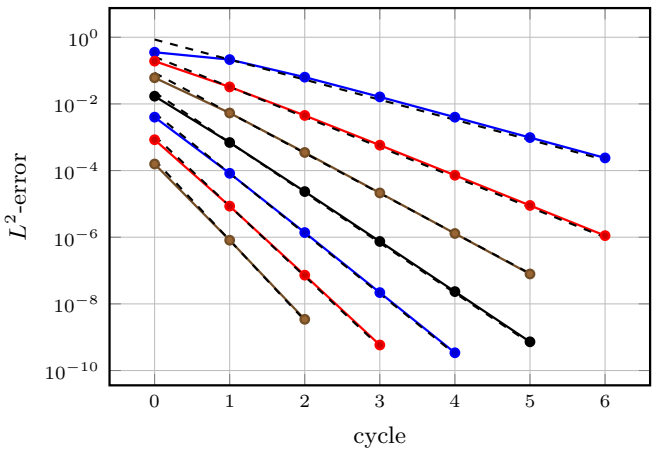
Continuous Galerkin, Sparse Matrix Vector



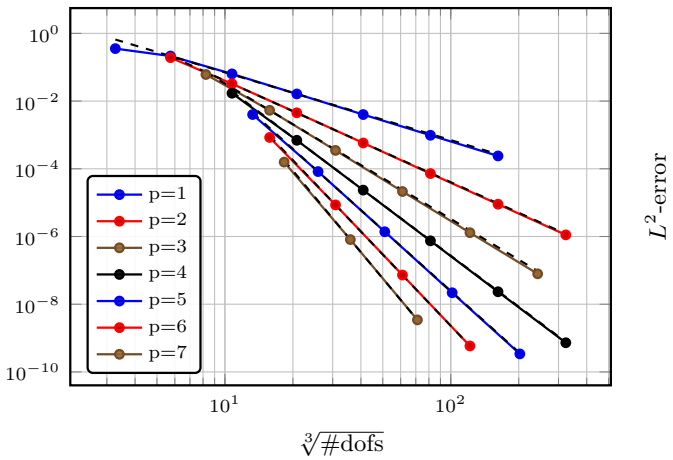
(a) Blend and Warp: error vs cycle



(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$



(c) Equidistant: error vs cycle

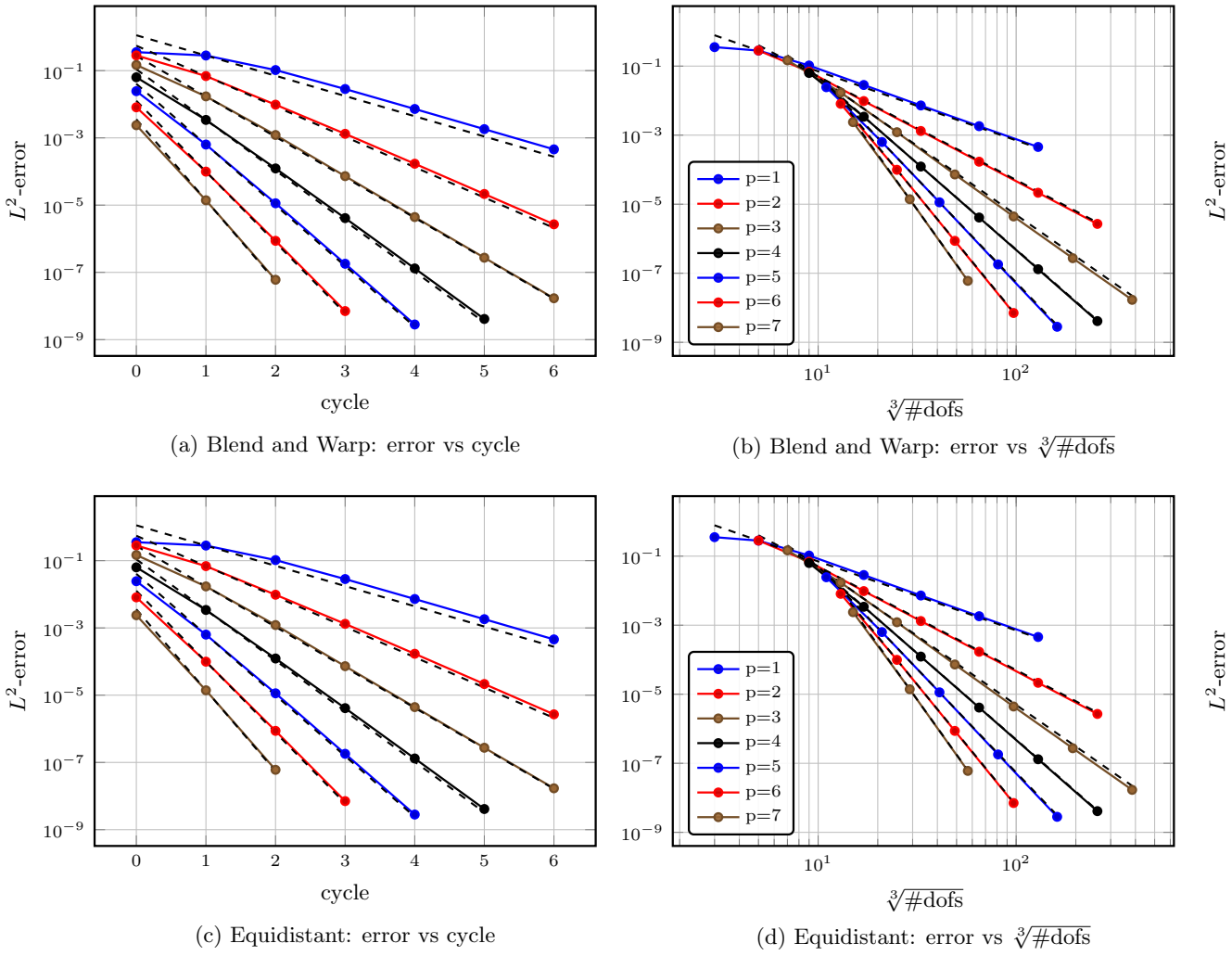


(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 21: MixedPyramid, Continuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes

MixedWedge

Continuous Galerkin, Sparse Matrix Vector



(a) Blend and Warp: error vs cycle

(b) Blend and Warp: error vs $\sqrt[3]{\#\text{dofs}}$

(c) Equidistant: error vs cycle

(d) Equidistant: error vs $\sqrt[3]{\#\text{dofs}}$

Figure 22: MixedWedge, Continuous Galerkin, Sparse Matrix Vector: Blend and Warp and Equidistant nodes